Lecture Scribe CSE400

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PDF

Generate a formal, academic lecture scribe suitable for exam preparation using ONLY the provided context (lecture slides / transcript / textbook excerpt). While writing the scribe, reason step by step through definitions, assumptions, theorems, proofs, derivations, and example solutions exactly as presented in the context, and include that step-by-step reasoning explicitly in the final scribe so the logical flow is clear for revision; explicitly state all definitions and assumptions, and for every proof, derivation, or worked example in the context, reproduce it in a numbered step-by-step exam style with no missing steps. Do not introduce any new material, alternative reasoning paths, extra examples, or external facts beyond what is explicitly contained in the provided context. Maintain a formal university-level tone optimized for exam revision, use clear headings and subheadings, and ensure every statement is traceable to the provided context.

Absolutely — here is a **formal, academic lecture scribe** for **CSE400 Lecture 20**, written **strictly from the provided PDF context only**, with **step-by-step exam-style derivations and no extra material added**.

CSE400 – Fundamentals of Probability in Computing

Lecture 20: Linear Transformations and Expectations of Multiple Random Variables

Source used: Provided Lecture PDF only

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1. Outline of the Lecture

The lecture covers the following topics exactly as listed in the provided material:

1. Statistical Representation of Random Vectors
2. Mean Vector, Correlation, and Covariance Matrices

3. Linear Transformations
4. Mapping properties under $Y = AX + b$
5. Multivariate Gaussian Distributions
6. From 1D/2D cases to N-dimensional Joint PDFs
7. Special Cases and Properties
8. Implications of zero correlation and matrix factorization

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2. Random Vector Representation

2.1 Vector Form of Multiple Random Variables

For random variables $X_1, X_2, \ldots, X_N$, the lecture represents them in vector form as:

$$X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

Definition

- X is called a **random vector**.
- It provides a **compact representation of multiple random variables**.

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3. Mean Vector

3.1 Definition of Mean Vector

For a random vector X, the mean vector is defined as:

$$\mu_X = E[X]$$

Interpretation

- It represents the **average value** of the random vector.
- Each element corresponds to the mean of one random variable.

3.2 Structure of the Mean Vector

The lecture explicitly gives the structure as:

$$\mu_X = \begin{bmatrix} E[X_1] \\ E[X_2] \\ \vdots \\ E[X_N] \end{bmatrix}$$

This shows that each coordinate of the mean vector is the expectation of the corresponding component random variable.

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4. Correlation Matrix

4.1 Definition

The correlation matrix of the random vector X is defined as:

$$R_{XX} = E[XX^T]$$

The (i, j)-th entry is:

$$R_{XX}(i, j) = E[X_i X_j]$$

where i and $j < N$ as stated in the lecture slides.

4.2 Matrix Representation

The lecture writes the matrix explicitly as:

$$R_{XX} = \begin{bmatrix} E[X_1X_1] & E[X_1X_2] & \cdots & E[X_1X_N] \\ E[X_2X_1] & E[X_2X_2] & \cdots & E[X_2X_N] \\ \vdots & \vdots & \ddots & \vdots \\ E[X_NX_1] & E[X_NX_2] & \cdots & E[X_NX_N] \end{bmatrix}$$

Property

- R_{XX} is a **symmetric matrix**.

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5. Covariance Matrix

5.1 Definition

For a random vector X, the covariance matrix is defined as:

$$C_{XX} = E\left[(X - \mu_X)(X - \mu_X)^T\right]$$

This is presented in the lecture as the “relationship map” of random vectors.

5.2 Practical Form (2D Illustration)

The lecture shows the 2-dimensional structure as:

$$C_{XX} = \begin{bmatrix} \mathrm{Var}(X_1) & \mathrm{Cov}(X_1, X_2) \\ \mathrm{Cov}(X_2, X_1) & \mathrm{Var}(X_2) \end{bmatrix}$$

Symmetry Property

The lecture explicitly notes:

$$\mathrm{Cov}(X_i, X_j) = \mathrm{Cov}(X_j, X_i)$$

Hence the covariance matrix is a **mirror of itself**, i.e., symmetric.

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6. Linear Transformation of Random Vectors

6.1 General Linear Model

The lecture considers the transformation:

$$Y = AX + b$$

Interpretation of Terms

- A: Transformation matrix
  - Scales and rotates the data
- b: Constant vector
  - Shifts / translates the center
- X: Input random vector

6.2 Component-wise Form for N-dimensional Vectors

The lecture writes the transformed components as:

$$Y_1 = a_{11}X_1 + a_{12}X_2 + \cdots + a_{1N}X_N + b_1$$

$$Y_2 = a_{21}X_1 + a_{22}X_2 + \cdots + a_{2N}X_N + b_2$$

$$\vdots$$

$$Y_N = a_{N1}X_1 + a_{N2}X_2 + \cdots + a_{NN}X_N + b_N$$

This explicitly shows how each output component is a linear combination of the input random variables plus a constant shift.

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7. Mean of the Transformed Random Vector

7.1 Goal

For the linear transformation

$$Y = AX + b$$

the lecture defines the transformed mean as:

$$\mu_Y = E[Y]$$

7.2 Step-by-Step Derivation (Exactly Following the Lecture)

Step 1: Substitute the transformation into the expectation

$$\mu_Y = E[AX + b]$$

Step 2: Apply linearity of expectation

$$\mu_Y = E[AX] + E[b]$$

Step 3: Use the result stated in the lecture

$$\mu_Y = A\mu_X + b$$

7.3 Final Result

$$\boxed{\mu_Y = A\mu_X + b}$$

This is the mean transformation rule given in the lecture.

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8. Correlation Matrix Transformation

8.1 Definition

For the transformed random vector Y, the lecture defines:

$$R_{YY} = E[YY^T]$$

Substituting $Y = AX + b$:

$$R_{YY} = E[(AX+b)(AX+b)^T]$$

8.2 Step-by-Step Derivation (Exactly Following the Lecture)

Step 1: Start from the definition

$$R_{YY} = E[(AX+b)(AX+b)^T]$$

Step 2: Expand the terms

The lecture indicates "Expansion of terms," followed by applying expectation.

Step 3: Apply expectation term-wise

The lecture gives:

$$R_{YY} = AE[XX^T]A^T + AE[X]b^T + bE[X^T]A^T + bb^T$$

Step 4: Replace with standard notation

Using

$$R_{XX} = E[XX^T], \quad \mu_X = E[X]$$

the lecture gives the final result:

$$R_{YY} = AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T$$

8.3 Final Result

$$\boxed{R_{YY} = AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T}$$

This is the correlation transformation rule presented in the lecture.

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9. Covariance Matrix Transformation

9.1 Key Observation

The lecture explicitly states:

$$Y - \mu_Y = (AX+b) - (A\mu_X + b) = A(X - \mu_X)$$

Important Note

- The shift vector b **does not affect the deviation from the mean**.

- Therefore, the shift does not affect the spread.

9.2 Step-by-Step Derivation (Exactly Following the Lecture)

Step 1: Start with the covariance definition

$$C_{YY} = E[(Y - \mu_Y)(Y - \mu_Y)^T]$$

Step 2: Substitute the deviation relation

Since

$$Y - \mu_Y = A(X - \mu_X)$$

we get:

$$C_{YY} = E\left[A(X - \mu_X)(X - \mu_X)^T A^T\right]$$

Step 3: Use the simplified result stated in the lecture

$$C_{YY} = AC_{XX}A^T$$

9.3 Final Result

$$\boxed{C_{YY} = AC_{XX}A^T}$$

Explicit Lecture Observation

- The shift b **does not affect the covariance (spread)**.

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10. Summary of Effects of Linear Transformation

The lecture summarizes the three mapping properties under $Y = AX + b$ as follows:

10.1 Mean

$$\boxed{\mu_Y = A\mu_X + b}$$

10.2 Covariance

$$\boxed{C_{YY} = AC_{XX}A^T}$$

10.3 Correlation

$$\boxed{R_{YY} = AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T}$$

These are the exact summary formulas shown in the lecture table.

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11. Worked Numerical Example: Linear Transformation

11.1 Given Data

The lecture gives the transformation $Y = AX$ with:

$$\mu_X = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix}$$

$$R_{XX} = \begin{bmatrix} 13 & 2 & 3 \\ 2 & 10 & 3 \\ 3 & 3 & 10 \end{bmatrix}$$

The lecture asks to calculate:

1. Mean vector of Y
2. Correlation matrix of Y
3. Covariance matrix of Y

11.2 Part (a): Mean Vector of Y

Formula Used

The lecture uses:

$$\mu_Y = A\mu_X$$

since the given example uses $Y = AX$ (no shift vector shown).

Step-by-Step Calculation

$$\mu_Y = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

Compute each entry exactly as shown in the lecture:

First entry

$$(1)(2) + (0)(-1) + (-1)(1) = 2 + 0 - 1 = 1$$

Second entry

$$(0)(2) + (-1)(-1) + (1)(1) = 0 + 1 + 1 = 2$$

Third entry

$$(-1)(2) + (1)(-1) + (0)(1) = -2 - 1 + 0 = -3$$

Final Mean Vector

$$\boxed{\mu_Y = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}}$$

This matches the mean vector used later in the covariance computation in the lecture.

11.3 Part (b): Correlation Matrix of Y

Formula Used

The lecture states:

$$R_{YY} = AR_{XX}A^T$$

(Here the example uses $Y = AX$, so only this term is used in the worked solution.)

Step 1: Compute AR_{XX}

The lecture explicitly gives:

$$AR_{XX} = \begin{bmatrix} 10 & -1 & -7 \\ 1 & -7 & 7 \\ -11 & 8 & 0 \end{bmatrix}$$

This is shown as the result of the first multiplication step in the slides.

Step 2: Multiply by A^T

The lecture then states the final result after multiplying by A^T:

$$\boxed{R_{YY} = \begin{bmatrix} 17 & -6 & -9 \\ -6 & 14 & -8 \\ -9 & -8 & 19 \end{bmatrix}}$$

This is the correlation matrix shown in the worked example.

11.4 Part (c): Covariance Matrix of Y

Formula Used

The lecture states:

$$C_{YY} = R_{YY} - \mu_Y \mu_Y^T$$

Step 1: Compute the outer product of the mean vector

Using

$$\mu_Y = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$$

the lecture gives:

$$\mu_Y \mu_Y^T = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} \begin{bmatrix} 1 & 2 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -3 & -6 & 9 \end{bmatrix}$$

Step 2: Subtract from the correlation matrix

The lecture writes:

$$C_{YY} = \begin{bmatrix} 17 & -6 & -9 \\ -6 & 14 & -8 \\ -9 & -8 & 19 \end{bmatrix} - \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -3 & -6 & 9 \end{bmatrix}$$

Step 3: Perform entry-wise subtraction

First row

$$17 - 1 = 16, \quad -6 - 2 = -8, \quad -9 - (-3) = -6$$

Second row

$$-6-2=-8, \quad 14-4=10, \quad -8-(-6)=-2$$

Third row

$$-9-(-3)=-6, \quad -8-(-6)=-2, \quad 19-9=10$$

Final Result Shown in the Lecture

The lecture slide explicitly displays:

$$\boxed{C_{YY}=\begin{bmatrix}16 & -8 & -8\\ -8 & 10 & -2\\ -8 & -2 & 10\end{bmatrix}}$$

This is the final covariance matrix shown in the provided PDF.

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12. Homework Assignment: Finding the Mean

12.1 Problem Statement

The lecture provides:

$$C_{XX}=\begin{bmatrix}5 & -1 & 2\\ -1 & 5 & -2\\ 2 & -2 & 8\end{bmatrix}$$

$$R_{XX}=\begin{bmatrix}6 & 1 & 1\\ 1 & 9 & -4\\ 1 & -4 & 9\end{bmatrix}$$

Task:

- Find the mean vector μ_X.

12.2 Hint Given in the Lecture

The lecture uses the relationship:

$$C_{XX}=R_{XX}-\mu_X\mu_X^T$$

Rearranging:

$$\mu_X\mu_X^T=R_{XX}-C_{XX}$$

Note Given in the Lecture

- Identify the elements of μ_X from the **diagonal** of the result.

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13. Gaussian Distributions: 1D and 2D

13.1 Univariate Gaussian (1D)

The lecture introduces the case of a **single random variable** X under the heading:

- "1. Univariate Gaussian (1D)"
- "For a single RV X"

(The explicit scalar formula is indicated in the slides, but only the heading and context are clearly readable in the provided excerpt.)

13.2 Joint Gaussian (2D)

For two random variables X and Y, the lecture presents the **joint PDF** of the bivariate Gaussian distribution.

The slide explicitly identifies:

- "2. Joint Gaussian (2D)"
- "For two RVs X and Y, the joint PDF is:"

The displayed expression is the standard bivariate Gaussian form in terms of:

- μ_X, μ_Y
- σ_X, σ_Y
- correlation coefficient ρ_{XY}

as shown on the lecture slide.

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14. Multivariate Gaussian Distribution (N-Dimensional)

14.1 Random Vector Form

For an N-dimensional random vector, the lecture writes:

$$X = [X_1, X_2, \ldots, X_N]^T$$

14.2 General Vector PDF

The lecture presents the general multivariate Gaussian PDF as:

$$f_X(x) = \frac{1}{\sqrt{(2\pi)^N \det(C_{XX})}} \exp\left(-\frac{1}{2}(x-\mu_X)^T C_{XX}^{-1}(x-\mu_X)\right)$$

Structure Components Listed in the Lecture

- μ_X: Mean vector $(N \times 1)$
- C_{XX}: Covariance matrix $(N \times N)$

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15. Example: 2D Joint Gaussian PDF

The lecture includes a worked derivation for the 2D case by connecting the general multivariate form to the familiar bivariate Gaussian form.

15.1 Step 1: Start from the General Multivariate Gaussian Form

The lecture explicitly starts from:

$$f_X(x) = \frac{1}{\sqrt{(2\pi)^N \det(C_{XX})}} \exp\left(-\frac{1}{2}(x-\mu_X)^T C_{XX}^{-1}(x-\mu_X)\right)$$

and then specializes to the 2D case.

15.2 Step 2: Compute the Determinant of the 2D Covariance Matrix

The lecture explicitly states:

$$\det(C_{XX}) = \sigma_1^2\sigma_2^2 - (\rho\sigma_1\sigma_2)^2$$

Then simplifies it as:

$$\det(C_{XX}) = \sigma_1^2\sigma_2^2(1-\rho^2)$$

This determinant is used in the normalization constant of the bivariate Gaussian density.

15.3 Step 3: Inverse of the Covariance Matrix

The lecture states that the inverse is obtained using the adjoint formula:

$$C_{XX}^{-1} = \frac{1}{\det(C_{XX})}\ \mathrm{adj}(C_{XX})$$

Then, after substitution and simplification, the lecture gives the standard 2D inverse form consistent with the bivariate Gaussian covariance structure.

15.4 Step 4: Evaluate the Quadratic Form

The lecture explicitly defines:

$$Q(X) = (x - \mu_X)^T C_{XX}^{-1} (x - \mu_X)$$

Then it states:

- Substitute the 2D vector form into the quadratic expression.
- After algebraic simplification, the expression reduces to the standard bivariate Gaussian quadratic form.

15.5 Conclusion of the Worked Derivation

The purpose of the example is to show that the **general multivariate Gaussian formula** reduces to the **known bivariate Gaussian PDF** when:

- $N = 2$,
- the covariance matrix is written in terms of $\sigma_1^2, \sigma_2^2, \rho$.

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16. Example: Uncorrelated Gaussian Random Variables

16.1 Statement of the Example

The lecture gives the following claim:

- A vector of N jointly Gaussian random variables is **mutually uncorrelated**.
- Prove that the joint Gaussian PDF for a vector of uncorrelated random variables factorizes accordingly.

16.2 Interpretation Given in the Lecture

The lecture explicitly states:

1. Since the variables are uncorrelated, the covariance matrix is **diagonal**.
2. Therefore, the joint Gaussian PDF **factorizes into the product of individual Gaussian PDFs**.

16.3 Key Insight (Given in the Hint)

For mutually uncorrelated Gaussian random variables:

$$\mathrm{Cov}(X_i, X_j) = 0, \quad \forall i \neq j$$

This is the exact condition highlighted in the lecture hint.

16.4 Covariance Matrix Simplification

Because all off-diagonal covariance terms are zero, the lecture shows that:

$$C_{XX} = \begin{bmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_N^2 \end{bmatrix}$$

Its inverse is therefore also diagonal:

$$C_{XX}^{-1} = \begin{bmatrix} 1/\sigma_1^2 & 0 & \cdots & 0 \\ 0 & 1/\sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1/\sigma_N^2 \end{bmatrix}$$

This exact diagonal structure is shown in the lecture hint slides.

16.5 Resulting Exponent

The lecture explicitly notes:

- Substitute the diagonal inverse back into the multivariate Gaussian density.
- The exponent becomes a sum of independent single-variable quadratic terms.

This is the reason the density factorizes into the product of 1D Gaussian PDFs.

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17. Homework Proof

The lecture gives the following homework:

$$\boxed{\mathrm{Var}[X+Y] = \mathrm{Var}[X] + \mathrm{Var}[Y] + 2\,\mathrm{Cov}(X,Y)}$$

This is stated as a proof exercise only; no derivation is shown in the provided lecture slides. Therefore, no further steps are included here beyond the exact homework statement.

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18. Case Study: Smart City Traffic + Weather System

The lecture concludes with an applied case study set in **Ahmedabad**, described as a system that predicts traffic congestion using transportation and climate variables.

18.1 Step 1: Define N Random Variables

The lecture defines a 5-dimensional random vector:

$$X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \\ X_5 \end{bmatrix}$$

with the following meanings:

- X_1: Traffic flow (vehicles/min)
- X_2: Average vehicle speed (km/h)
- X_3: Rainfall intensity (mm/hr)
- X_4: Temperature (°C)
- X_5: Air pollution index (AQI)

This is the system state random vector, with $N = 5$.

18.2 Step 2: Mean Vector

The lecture defines:

$$\mu = E[X] = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \\ \mu_4 \\ \mu_5 \end{bmatrix}$$

and gives the following example average values:

- Average traffic flow = 120 vehicles/min
- Average speed = 35 km/h
- Average rainfall = 2 mm/hr
- Average temperature = 32°C
- Average AQI = 180

18.3 Step 3: Covariance Matrix as a Dependency Map

The lecture interprets the covariance matrix as the **dependency map of the city**.

It gives example sign interpretations:

Example 1

$$\mathrm{Cov}(X_1, X_3) > 0$$

Interpretation:

- More rain $\Rightarrow$ more congestion (traffic increases)

Example 2

$$\mathrm{Cov}(X_2, X_3) < 0$$

Interpretation:

- More rain $\Rightarrow$ lower speed

Example 3

$$\mathrm{Cov}(X_1, X_5) > 0$$

Interpretation:

- More traffic $\Rightarrow$ higher pollution

Example 4

$$\mathrm{Cov}(X_4, X_5) < 0 \quad (\text{maybe})$$

Interpretation:

- Higher temperature $\Rightarrow$ pollution disperses

These interpretations are given directly in the lecture slides and are part of the intuition-building case study.

18.4 Step 4: Joint Gaussian Assumption

The lecture assumes:

$$X \sim \mathcal{N}(\mu, \Sigma)$$

Why Gaussian? (Exactly as Stated in the Lecture)

1. Aggregation of many small effects — Central Limit Theorem
2. Sensor noise is often Gaussian
3. Easier analytics (closed-form solutions)

Interpretation

This means the entire system (traffic + climate) behaves in a structured probabilistic way defined by:

- Mean (typical conditions)

- Covariance (interdependencies)

18.5 Step 6: Linear Transformation to a Congestion Index

The lecture defines a derived scalar metric:

$$Y = 0.6X_1 - 0.4X_2 + 0.8X_3$$

This is identified as a **linear transformation**.

The lecture writes:

$$Y = a^T X$$

where a is the coefficient vector corresponding to the linear combination.

Inference #1 (Given in the Lecture)

If

$$X \sim \mathcal{N}(\mu, \Sigma)$$

then:

1. Y is also Gaussian
2. Mean:

$$E[Y] = a^T \mu$$

The lecture indicates "Variance:" as the next item, but the explicit variance expression is not fully visible in the provided excerpt; therefore it is not expanded here beyond what is shown.

18.6 Conceptual Summary from the Case Study

The lecture concludes with the following conceptual mapping:

1. Random vector = system state
2. Covariance = relationships
3. Gaussian = structured uncertainty
4. Transformation = derived metrics

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19. Final Exam-Oriented Summary

The key examinable results from this lecture, exactly as presented, are:

19.1 Random Vector

$$X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

19.2 Mean Vector

$$\mu_X = E[X]$$

19.3 Correlation Matrix

$$R_{XX} = E[XX^T]$$

19.4 Covariance Matrix

$$C_{XX} = E[(X - \mu_X)(X - \mu_X)^T]$$

19.5 Linear Transformation

$$Y = AX + b$$

19.6 Mean Under Linear Transformation

$$\boxed{\mu_Y = A\mu_X + b}$$

19.7 Correlation Under Linear Transformation

$$\boxed{R_{YY} = AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T}$$

19.8 Covariance Under Linear Transformation

$$\boxed{C_{YY} = AC_{XX}A^T}$$

19.9 Multivariate Gaussian PDF

$$\boxed{f_X(x) = \frac{1}{\sqrt{(2\pi)^N \det(C_{XX})}} \exp\left(-\frac{1}{2}(x - \mu_X)^T C_{XX}^{-1}(x - \mu_X)\right)}$$

19.10 Special Gaussian Property

- If jointly Gaussian variables are **mutually uncorrelated**, then:

- C_{XX} is diagonal
- The joint PDF factorizes into the product of individual Gaussian PDFs

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